

MatchMind AI

Multi-Agent Cricket Intelligence Platform

Academic Project Documentation

1. Project Overview

MatchMind AI is a domain-specific multi-agent cricket intelligence platform. It retrieves cricket match and scorecard information through the Cricbuzz API, converts the response into structured match context, sends the context to multiple specialist AI agents, and synthesizes their findings into a final decision-support report.

2. Objectives

- Build a practical multi-agent AI system for cricket analysis.
- Separate analysis into focused specialist roles instead of relying on one general-purpose agent.
- Integrate external cricket data with structured parsing and orchestration.
- Produce a concise, actionable final report while maintaining data-grounding constraints.

3. System Architecture

The application follows a modular pipeline. The Streamlit interface accepts the selected match, the engine retrieves data through the API client, and the parser creates structured match context. The Agent Manager/Orchestrator then executes five specialist agents. Their outputs are passed to a dedicated Report Generator Agent.

MatchMind AI — Multi-Agent Cricket Intelligence Architecture

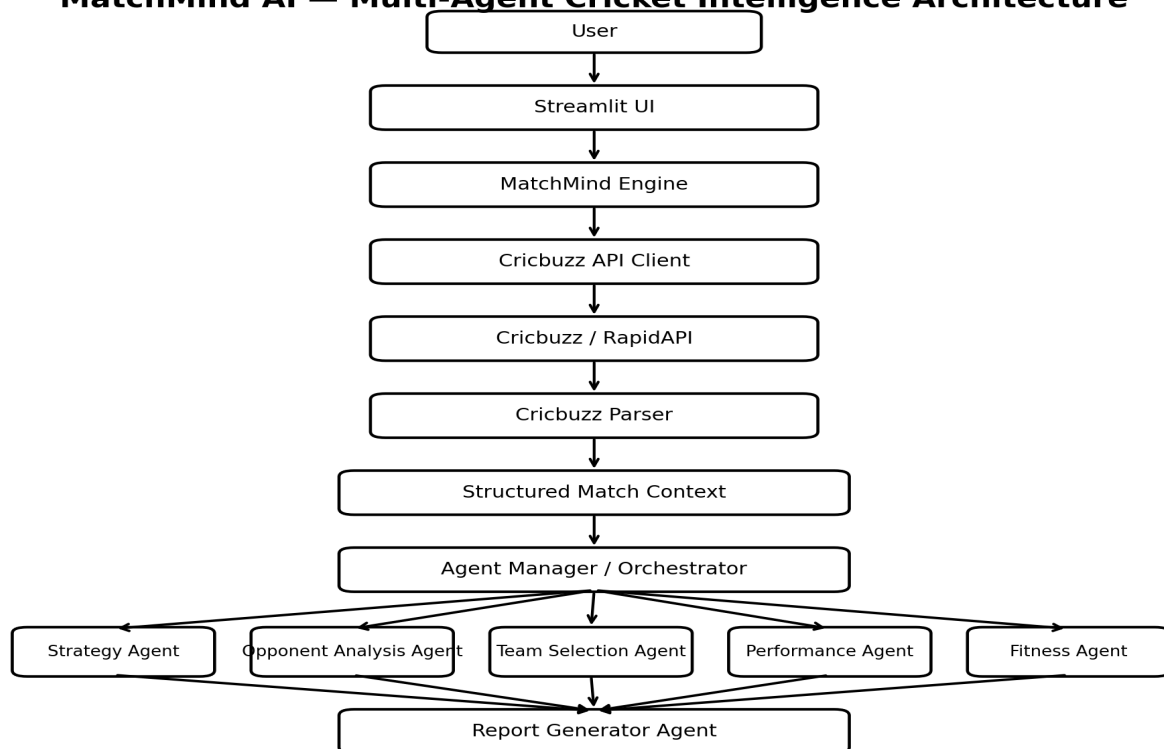


Figure 1. MatchMind AI system architecture.

4. Multi-Agent Components

Agent	Primary Responsibility
Strategy Agent	Tactical and strategic match analysis
Opponent Analysis Agent	Opponent strengths, weaknesses and insights
Team Selection Agent	Squad and player-selection insights
Performance Agent	Batting and bowling performance analysis
Fitness Agent	Workload and fitness-related assessment
Report Generator Agent	Synthesizes specialist outputs into the final report

5. Analysis Workflow

1. User selects a cricket match in the Streamlit interface.
2. Match and scorecard data are retrieved from Cricbuzz through RapidAPI.
3. The parser transforms the API response into structured match context.
4. The orchestrator determines and executes the required specialist agents.
5. Specialist outputs are collected and passed to the Report Generator Agent.
6. The final report is rendered in the Streamlit interface.

6. Technology Stack

Programming: Python **UI:** Streamlit **Agent Framework:** CrewAI
Data: Cricbuzz / RapidAPI **HTTP:** Requests **Configuration:** python-dotenv
Version Control: Git and GitHub

7. Reliability and Data-Grounding

Each agent execution is protected by a 300-second timeout. If an individual agent fails or times out, the orchestrator records the issue and can continue using completed specialist outputs. The Report Generator is instructed to use only the supplied match context and specialist reports, explicitly identify unavailable information, and avoid fabricating statistics, player information, injuries, workloads, or match events.

8. Output and Evaluation

The implemented system successfully produces an interactive match view containing match details and scorecard information, followed by specialist-agent execution status and a structured final MatchMind AI report. The report is organized into Match Overview, Executive Summary & Strategy, Squad & Opponent Insights, Fitness & Workload Assessment, and Key Recommendations & Data Limitations.

9. Conclusion

MatchMind AI demonstrates a practical application of multi-agent AI in a specialized domain. The project combines external API integration, structured data processing, role-based agent reasoning, orchestration, reliability controls, and an interactive user interface into one coherent academic prototype.

Project Repository: github.com/kss1510/MatchMind-AI